

AVIONICS TASK ROUND 2

TASK-1

BUILD 1

USE CASE:

Build-1 is the best option out of all as even though depth cameras can detect the object easily compared to a stereo one, it fails to do the job outdoors as the sunlight can impact its infrared rays therefore decreasing its object detection capability and battery life.

COMPUTE:

For computing purposes it has nvidia jetson orino which manages this part thereby reducing the computing load to some extent.

POWER:

The power consumption is found to be around 10-15 watts which might be pretty high even though manageable.

COST:

The nvidia flight computer costs relatively higher than raspberry Pi but since it fulfills the computational demands this is generally recommended

WEIGHT:

It is slightly heavier than the raspberry pie but not a very concerning factor

DECREASING ORDER OF PRIORITY OF ALL CAMERAS

Build 1 nvidia gpu better for low computation load

Build 4 raspberry pie so all computation load goes on cpu
(too much strain)

Build 5 can handle sunlight exposure but not to the extent of stereo cameras

Build 2 - fails due to sunlight exposure

Build 3 - fails due to sunlight exposure

BONUS QUESTION

a)The drone while moving forward might tilt a bit downwards hence the cameras infront of the drones can be tilted 15 degrees upwards so that we can minimise the error.

b)About the alternate build my suggestion would be is to construct a drone that is resistant to sunlight so that indepth cameras can be used to its fullest as we are not able to exercise the complete advantages of the in depth cameras just because of the sunlight factor.

TASK-2

RC transmitter and receiver:

Used to establish the connection between controller and the drone. Transmits the flight commands given on the controller to the drone. For Transmitter **RadioMaster Boxer** and for Receiver **RadioMaster RP3 ExpressLRS (ELRS) 2.4GHz** is the best choice as it easily bypasses PWM/PPM mismatches through high speed digital serial protocol (CRSF). can connected by connecting the receiver's TX, RX, 5V, and GND pins to one of the available UART serial ports on the Pixhawk 6C Mini. It is Powered directly from the 5V regulated output pin of the Pixhawk's serial/RC connector rail.Communicates with the flight controller using a digital protocol called **CRSF (Crossfire)**. ELRS has incredible range, easily reaching 5 to 10+ kilometers line-of-sight.

Telemetry Module:

Its like a wireless USB cable that transmits real time flight data such as GPS coordinates,speed of the drone etc,can also be used to change the drones automatic flight path. **Holybro SiK Telemetry Radio** is the best model for the given use statement.Plugs straight into the "TELEM 1" or "TELEM 2" port on the Pixhawk.Powered by the Pixhawk's 5V TELEM port.it speaks a standard robotics language called **MAVLink**. Depending on the antennas, the range is typically 1 to 3 kilometers.

VTX:

Transmits real time drone footage from drone to camera to the flight monitor or the FPV(First person view) goggles. Digital VTX like DJI is a better choice compared to analog VTX because even though digital ones produce a slight delay and are expensive they provide HD quality video which is very useful for the given task. We can connect the video signal wire running from the camera to the VTX.

It is generally recommended to not use Pixahawk for powering the VTX as it draws a lot of current instead we can use the main drone battery to power it. Digital VTX uses proprietary encoded digital signals over 2.4GHz bands, the video is compressed into data packets, providing a crystal-clear 720p or 1080p HD feed. Digital signals have far better range than the analog ones they can send signals from 4-15 Kms

DRONE COMMUNICATION DIAGRAM

<https://canva.link/7k5phd2o0c2yoyu>

P.S The arrows I haven't named transport the given commands through wireless signals

TASK-3

BATTERY:

4S 5200mAh LiPo Battery is the best choice that adheres to the given requirements also keeping the drone weight under 2.5 kgs

The assumed weight is around 2kg which leaves 500g for any error. The thrust required to hover the quadcopter is around 500g per motor.

The Holybro 2216 920KV motor yields approximately 7 g/W efficiency at a 500g hover load

Hence Total Hover power = $2000\text{g}/7\text{g/watt} = 285.6\text{W}$

Hover current = $285.6/14.8V = 19.3A$

For Endurance to fly for 15 minutes just in case for a safe buffer over 12 minutes $19.3A \times 0.25h = 4.825mAh(4825mAh)$

Since the recommended battery is a 5200mAh it safely satisfies the given requirements

ESC BUILD:

A **40A 4-in-1 ESC SpeedyBee BLHeli_S 40A** is the preferred esc build

The motors only use a maximum of 18 Amps. Because the ESC can safely handle 40 Amps, it has more than double the capacity it actually needs.

The massive safety buffer means the electronics will stay cool and safe, even when you push the throttle to the maximum or fight strong winds.

Long back builders used to mount four separate, bulky ESC circuit boards on the arms of the drone. Now, A 4-in-1 ESC packs all four into one small, square board that sits cleanly in the center of the frame, saving weight and hiding the wires

BONUS:

BrotherHobby Avenger 2806.5 (870KV) on a 6S battery.

The provided 2216 stator is relatively narrow for a 10-inch propeller, meaning the motor wastes energy as heat attempting to maintain the necessary torque. Upgrading to a wider motor drastically increases mechanical leverage, swinging the 1045 blades effortlessly. Pairing this larger motor with a higher voltage 6S battery (22.2V) drops the electrical current requirement, pushing hover efficiency toward 9 g/W–10 g/W. This configuration generates significantly higher absolute thrust consuming less power.